

Alvaro Guerra

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EDUCATION

Texas A&M University

B.S. Mechanical Engineering

College Station, TX

2021 – 2025

EXPERIENCE

Structures Lead

11-2024 – 06-2025

IGNITORS Rocketry Team

College Station, TX

- Led structural design of high-altitude COTS rocket; sized fiberglass airframe components using SolidWorks, ANSYS, and hand calculations with FOS >2
- Created Excel BOMs, fabrication drawings, and purchase orders for composites and hardware material, delivering 1% under \$3.5k budget
- Collaborated across avionics and propulsion teams to resolve fit, clearance, and structural handoff issues under tight deadlines

Structures and Materials Designer

05-2022 – 04-2023

SAE Aero Design Team

College Station, TX

- Modeled and analyzed 18-ft span cargo aircraft using SolidWorks and ANSYS to optimize payload and meet field assembly constraints
- Applied DFM principles by creating manufacturing jigs used to simplify the assembly process for aircraft fuselage, empennage, and control surfaces
- Led rapid field repair under competition pressure—enabling last flight that secured 1st place internationally among 40+ teams

Capstone Design Engineer

08-2024 – 05-2025

Garden Cultivation Robot Project

College Station, TX

- Integrated hydraulic pump actuation with Arduino/Pi electronics to automate watering, improving water efficiency by 26%
- Diagnosed drivetrain failure from loose chain and firmware issues, restoring full mobility for final testing and production release
- Machined chassis and custom mounts using CNC and 3D printing, delivering all design goals \$200 under budget

Real Estate Analyst Intern

05-2021 – 08-2021

River Valley Appraisals

Edinburg, TX

- Built Excel regression models to improve pricing accuracy on over 100+ residential properties

PROJECTS

BeamMeUpScotty (Python, Jupyter)

April 2025

- Developed 2-D beam/truss solver using Euler-Bernoulli theory and Newton-Raphson iteration to output GUI and interactive diagrams for fast validation of structural assumptions

TECHNICAL SKILLS

Software: Siemens NX, SolidWorks, Onshape, ANSYS Mechanical, Python (*pandas, NumPy, Matplotlib, PyQt*), MATLAB, HTML/CSS, Git, Microsoft Office

Manufacturing: Composite Layups, 3D Printing (Prusa/Ender3), CNC Machining, Manual Mill/Lathe, Hand Tools

Additional: English/Spanish Fluency, DFM/FMEA, Agile Methodology, Lean Philosophy